

Welcome All

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Mastering MySQL®
(A Placement Oriented Course)

Presented By:
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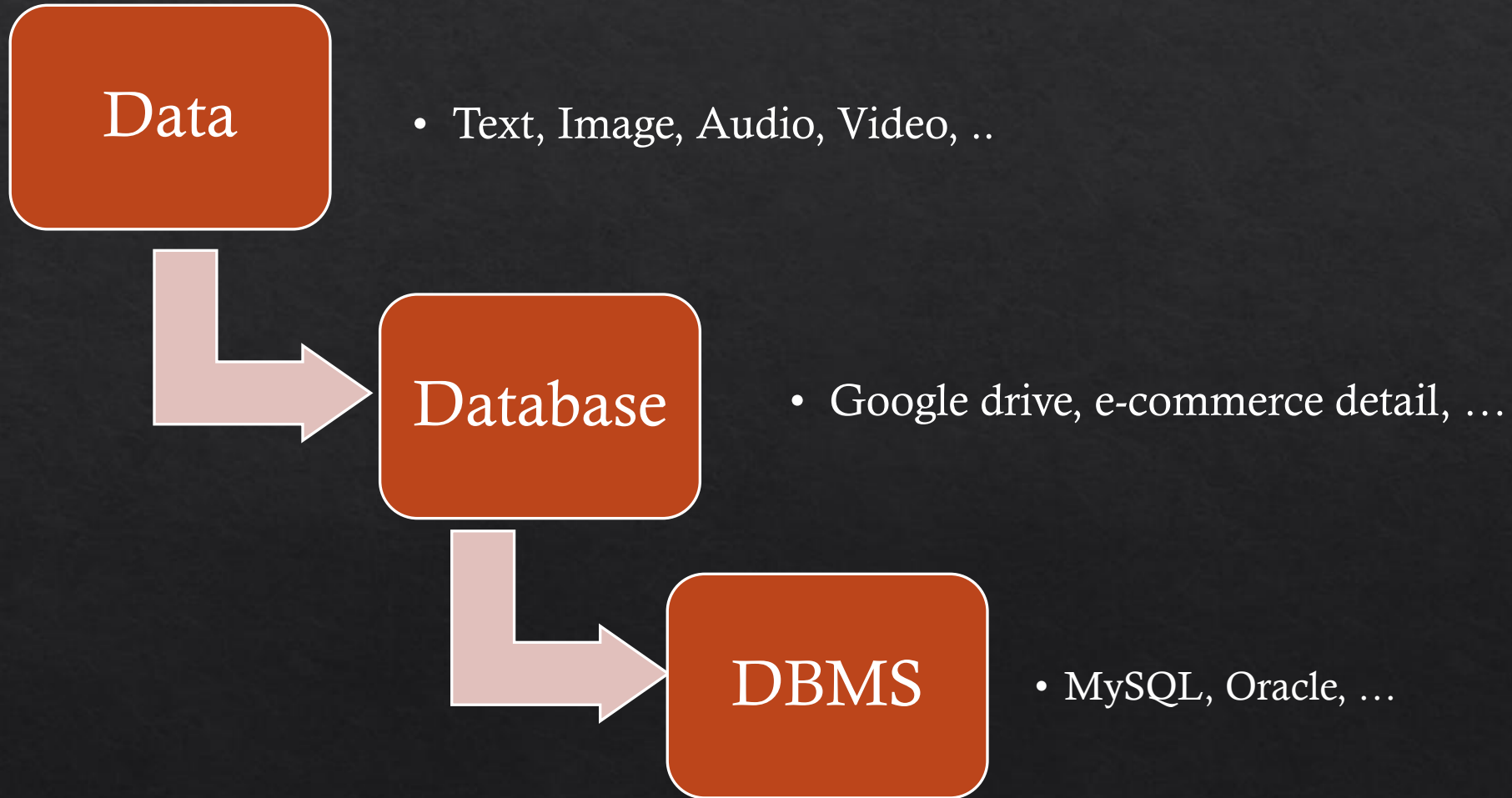
What is DBMS?

Def: *A DBMS is software used to : Store data, Schema design, Manage data, Retrieve data efficiently from database.*

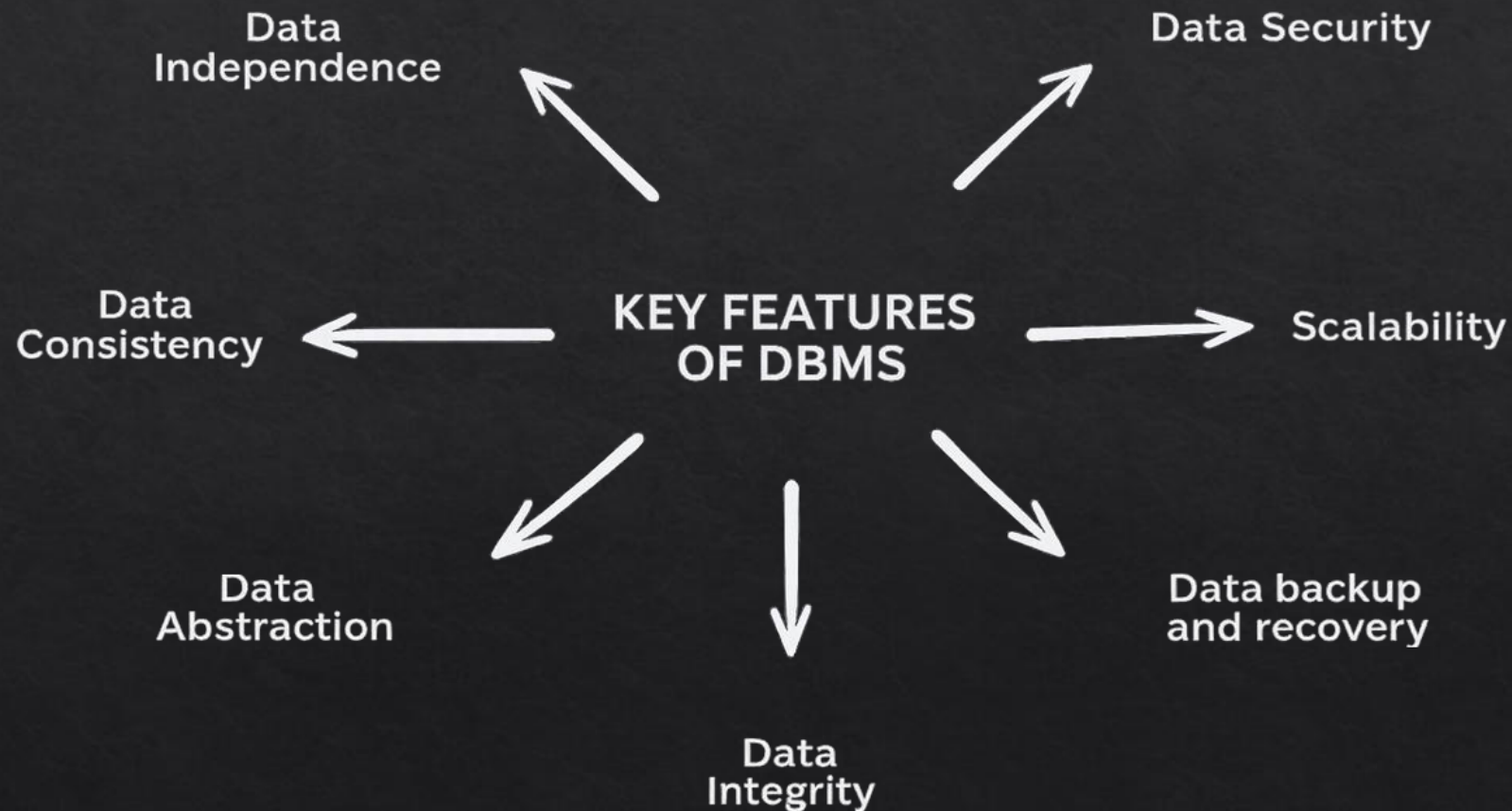
◇ DBMS acts as a bridge between: **User & Database**



Data vs Database vs DBMS



Key features of DBMS



DBMS vs File System

- A File System stores data in normal files and folders.
- A DBMS (Database Management System) stores and manages data in an organized database.



Examples of DBMS



Evolution of DBMS

DBMS

```
graph TD; DBMS[DBMS] --- Flat[Flat file database]; DBMS --- RDBMS[RDBMS]; DBMS --- NoSQL[NoSQL]; Flat --- FlatText[Text editors or Spreadsheets can be used to create flat file database]; RDBMS --- RDBMSList[Oracle, MySQL, IBM DB2, MS SQL Server, SAP ASE]; NoSQL --- NoSQLList[MongoDB, Oracle NoSQL Database, Apache Cassandra DB];
```

Flat file database

Text editors or Spreadsheets
can be used to create flat file database

RDBMS

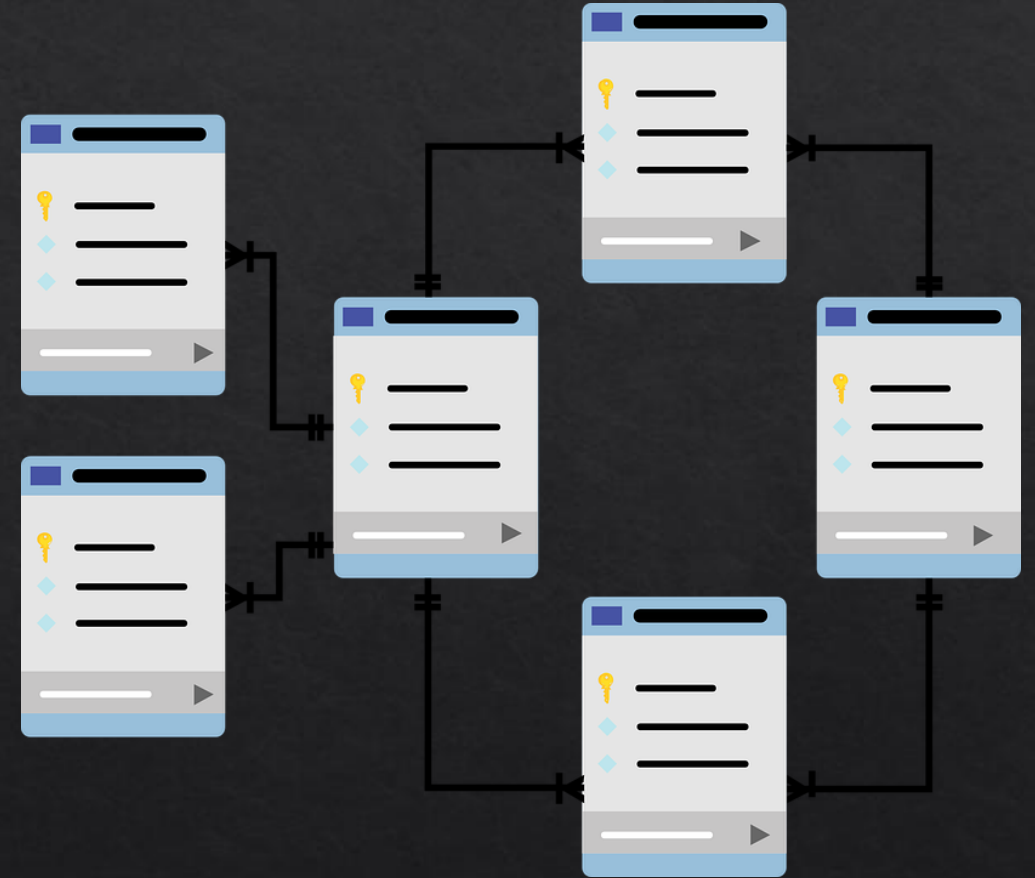
Oracle, MySQL, IBM DB2,
MS SQL Server, SAP ASE

NoSQL

MongoDB, Oracle NoSQL Database,
Apache Cassandra DB

Introduction to RDBMS

- **RDBMS** stands for Relational Database Management System.
- All modern database management systems like SQL, MS SQL Server, IBM DB2, ORACLE, MySQL, and Microsoft Access are based on RDBMS.
- RDBMS is a type of DBMS in which data is stored in the form of Relations/Tables.



What is a Relation/Table

- The RDBMS database uses tables to store data.
- A table is a collection of related data entries and contains rows and columns to store data.

EMPLOYEE				DEPARTMENT	
EMP ID	EMP_NAME	EMP_ADD	DEP_ID	DEP_ID	DEP_NAME
01	JOE	CAPE TOWN	D_101	D_101	ECO
02	ANNIE	FRANSISCO	D_103	D_102	CS
03	PETER	CROY CITY	D_101	D_103	JAVA
04	JOHN	FRANSISCO	D_102	D_104	MATHS
05	LUNA	TOKYO	D_106	D_105	BIO
06	SONI	W.LAND	D_105	D_106	CIVIL
07	SAKACHI	TOKYO	D_104		
08	MARY	NOVI	D_101		

Structure of a Relation

Unique relation name

Attributes or Fields or

Row, Record or Tuple

EMPLOYEE			
EMP_ID	EMP_NAME	EMP_ADD	DEP_ID
01	JOE	CAPE TOWN	D_101
02	ANNIE	FRANSISCO	D_103
03	PETER	CROY CITY	D_101
04	JOHN	FRANSISCO	D_102
05	LUNA	TOKYO	D_106
06	SONI	W.LAND	D_105
07	SAKACHI	TOKYO	D_104
08	MARY	NOVI	D_101

Degree (No. of Columns)=4

Cardinality (No. of Rows)=8

Properties of a Relation

- No two relation have same name
- Relation does not contain duplicate rows/tuples
- All tuples of the relation have the same format and the same number of entries.
- Null values are permitted.
- In relation, each cell contain exactly one value.

EMPLOYEE				DEPARTMENT	
EMP ID	EMP_NAME	EMP_ADD	DEP_ID	DEP_ID	DEP_NAME
01	JOE	CAPE TOWN	D_101	D_101	ECO
02	ANNIE	FRANSISCO	D_103	D_102	CS
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07	SAKACHI	TOKYO	D_104		
08	MARY	NOVI	D_101		

DBMS Architectures

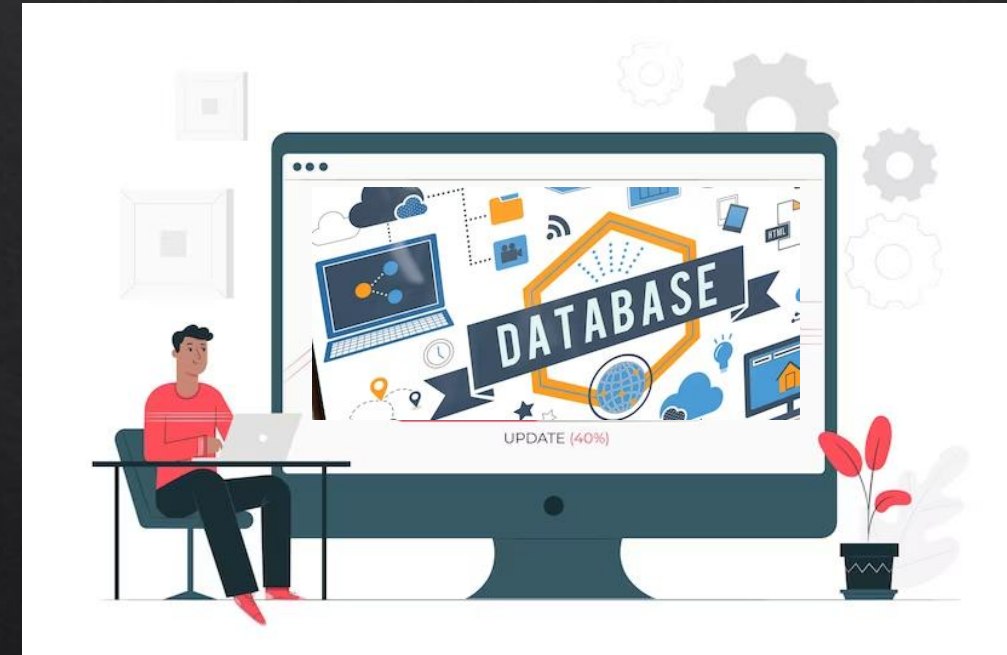
- DBMS architecture is based upon how the user is connected with specific database system.
- DBMS architecture decide the overall structure of how actually user will be able to access the data from the database.
- DBMS architecture effects the performance(access, maintain and security) of the database.

Types of DBMS Architectures:

- 1- Tier Architecture
- 2- Tier Architecture
- 3- Tier Architecture

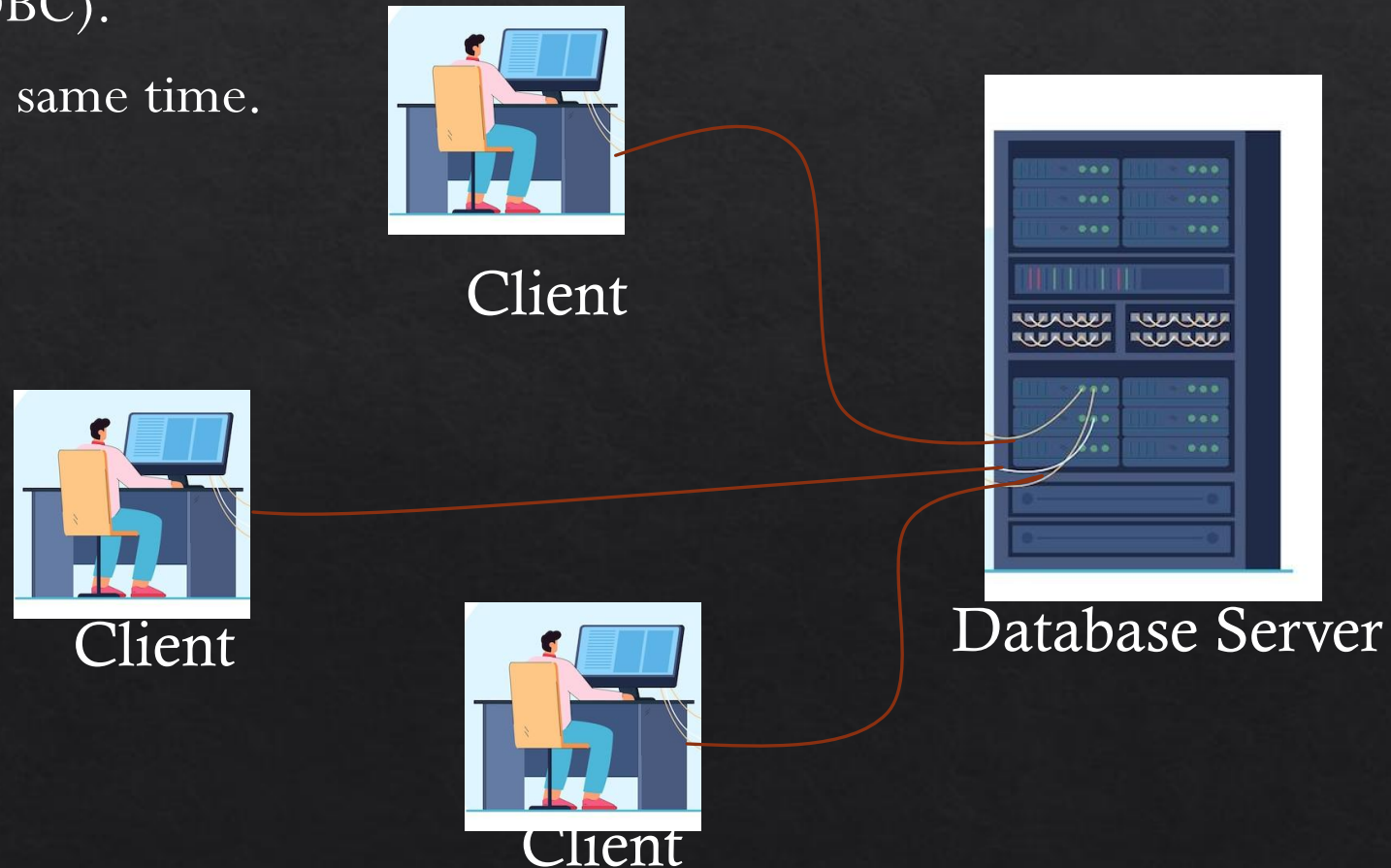
1- Tier Architecture

- All the DBMS components reside on a single server or platform.
- The database is directly accessible by the end-user. Means, any changes done here will directly be done on the database itself.
- Client, server, and Database all reside on the same machine.
- No multiple users can access database at the same time.
- The 1-Tier architecture is used for development of the local application



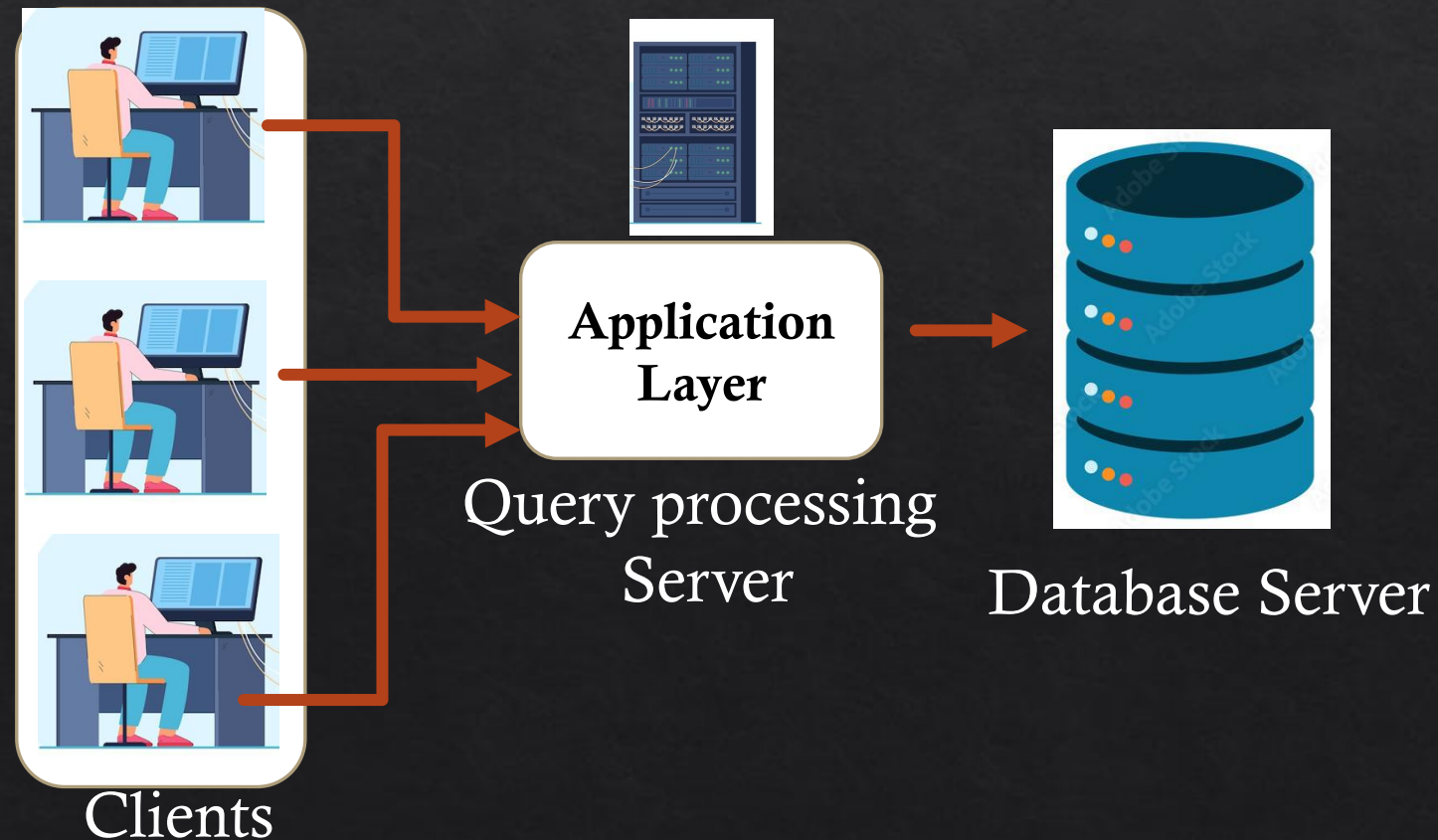
2-Tier Architecture

- 2-Tier architecture is similar to Client-Server architecture.
- Database reside on the server.
- The clients can access the database stored on the server by using API's (application programming interfaces, e.g. ODBC, JDBC).
- Multiple users can access database at the same time.
- Has scalability and Security issues.



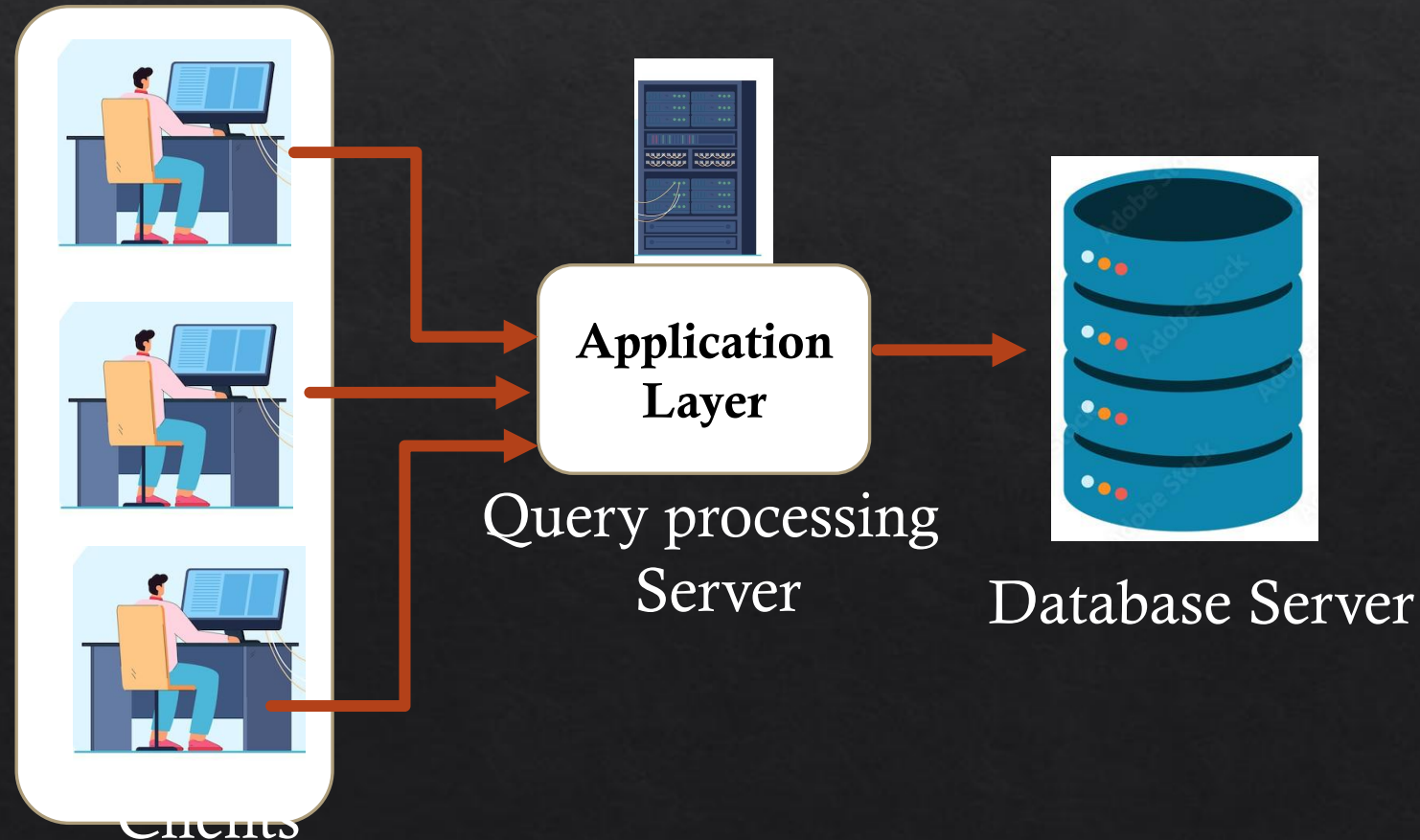
3-Tier Architecture

- In 3-Tier architecture another layer known as Intermediate or Application layer is added between the server (Database Layer) and the client (Presentation Layer) systems.
- There is no direct connection between the client and the Server.
- The intermediate layer reduce the query processing load of the server and improve security.



Working of 3-Tier Architecture

- All the Clients requests are handled by the Application layer server, i.e., the requests sent by the users are checked and verified by the Intermediate/ Application Layer server before transferring them to the database server.
- The intermediate layer provides the end-user an abstract view of the database.



Database Schema representation

- Database schema is the blueprint of how the database is constructed.
- Database access, creation and modification become easy for the “end user” and “programmer” with the help of schema.
- A database schema represents the logical structure of the database.



User View



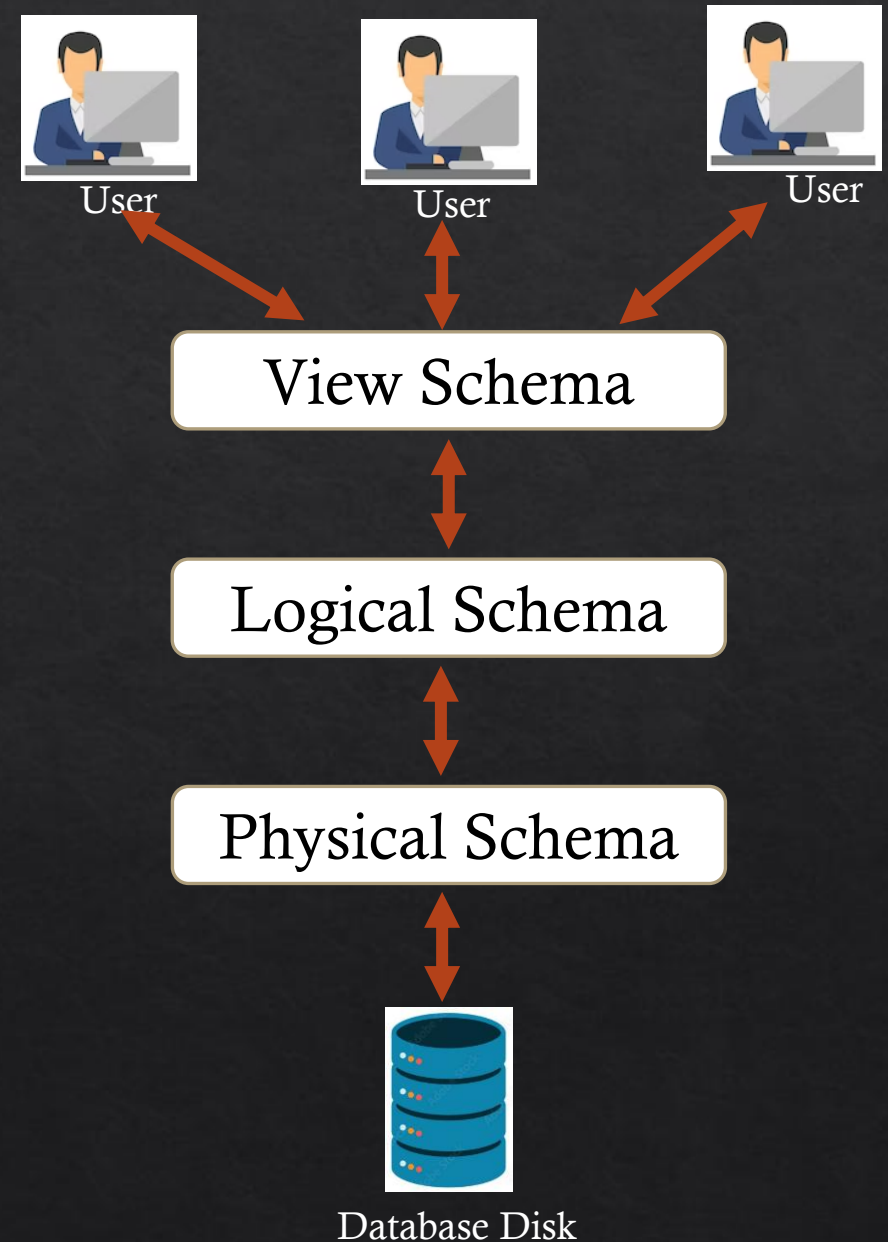
Architect View



Physical View

Three Schema Architecture

- There are three types of database schemas:
 1. View schema or External schema
 - Decide how data will be visible to user.
 2. Logical schema or Conceptual schema
 - Describe the attributes/columns and other details of the relation.
 3. Physical schema or Internal schema
 - Describes how the data is actually/physically present on hardware.



Other Important Points about Database Schema

- Makes easier for end user to insert, update or modify the data without going in logical and physical detail of data storage.
- Programmers works at logical level and perform modification without worrying about how database will be store on physical media.
- The modifications at one level can be made without worrying about other levels.
- Once the schema is defined and database is created, it is very difficult to modify/update schema later.
- Schema does not represent any data stored in the database. The data which is present in the database at a particular time is known as **Instance** in DBMS

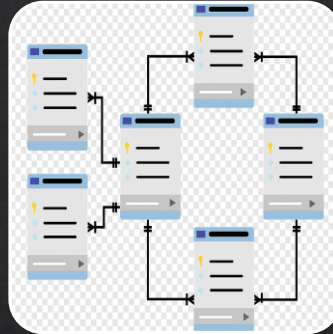
Database design process



Requirement
gathering



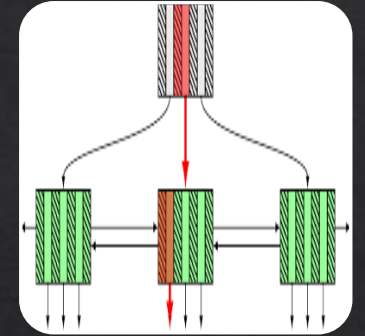
E-R
diagrams



Relational
schema



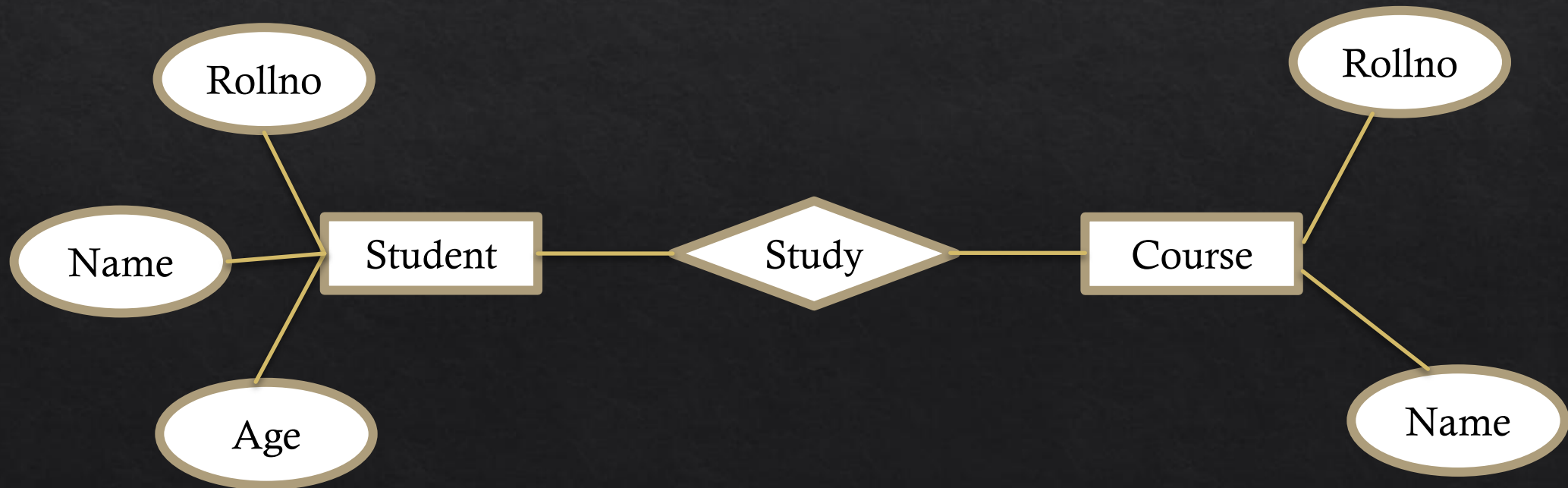
Normalize
conceptual
schema



Physical
schema

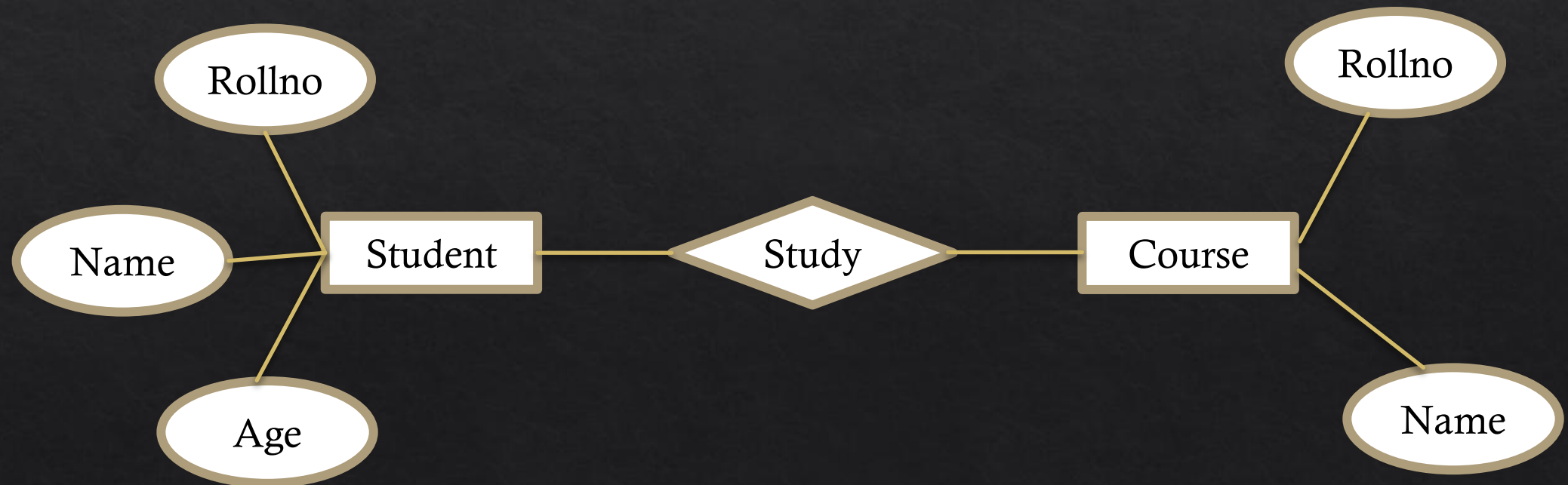
Introduction to ER Model

- Entity Relationship model represents “High level” database design.
- ER Model helps to finalize database design demanded by user in the form on ER diagrams.
- ER Model makes it easy to design database for database designer.



Components of ER Model

- ER model have 3 main components:
 - I. Entity
 - II. Relationship
 - III. Attributes



Introduction to MySQL

Definition: MySQL is an open-source **Relational Database Management System (RDBMS)** used to store, organize, manage, and retrieve data efficiently.

- It uses **SQL (Structured Query Language)** to communicate with databases.
- Features of MySQL:
 - ✓ Open Source
 - ✓ Fast performance
 - ✓ Secure
 - ✓ Multi-user support
 - ✓ Supports huge databases
 - ✓ Cross-platform
- MySQL Workbench is a graphical tool for MySQL, which is used for: Writing SQL queries, Creating tables, Database design, Administration.

Installation of MySQL

Search “mysql installer for windows” in google.

mysql installer for windows - Google

google.com/search?q=mysql+installer+for+windows&sca_esv=902c31b61ac8c1b2&xsrf=ANbL-n5FhwVcz_02SeCS3q6MqH0Rid5sIg%3A1778601448740&source=hp&ei=6...

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MySQL Installer 5.7.44
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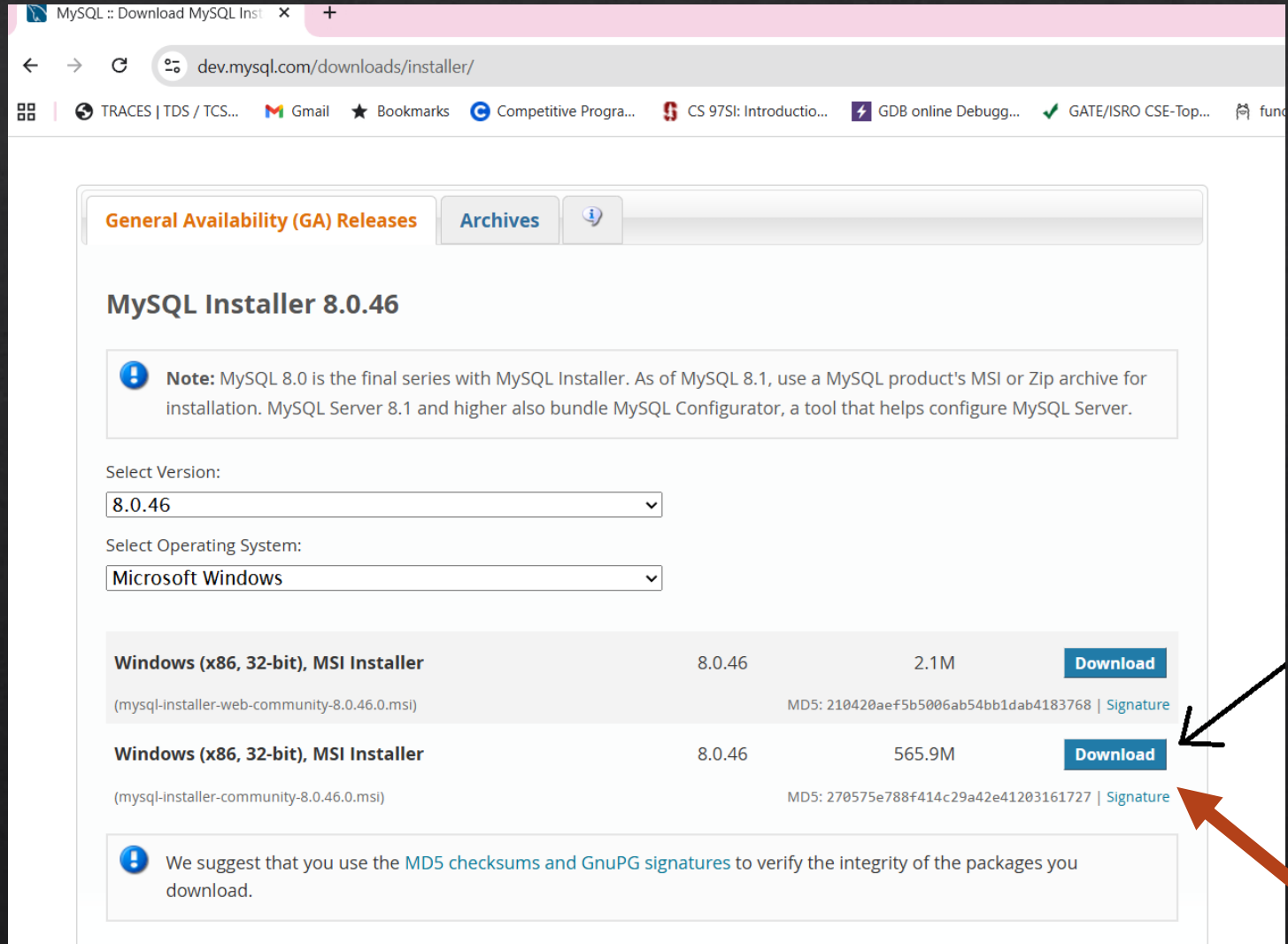
Videos

How To Install MySQL on Windows 11 (2026)

YouTube · ProgrammingKnowledge
11 Jan 2026

Learn to **install MySQL** Community Server and **MySQL Workbench** on **Windows 11**, including setup, configuration, and connecting.

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MySQL :: Download MySQL Installer

dev.mysql.com/downloads/installer/

General Availability (GA) Releases Archives

MySQL Installer 8.0.46

Note: MySQL 8.0 is the final series with MySQL Installer. As of MySQL 8.1, use a MySQL product's MSI or Zip archive for installation. MySQL Server 8.1 and higher also bundle MySQL Configurator, a tool that helps configure MySQL Server.

Select Version:
8.0.46

Select Operating System:
Microsoft Windows

Windows (x86, 32-bit), MSI Installer (mysql-installer-web-community-8.0.46.0.msi)	8.0.46	2.1M	Download
Windows (x86, 32-bit), MSI Installer (mysql-installer-community-8.0.46.0.msi)	8.0.46	565.9M	Download Signature

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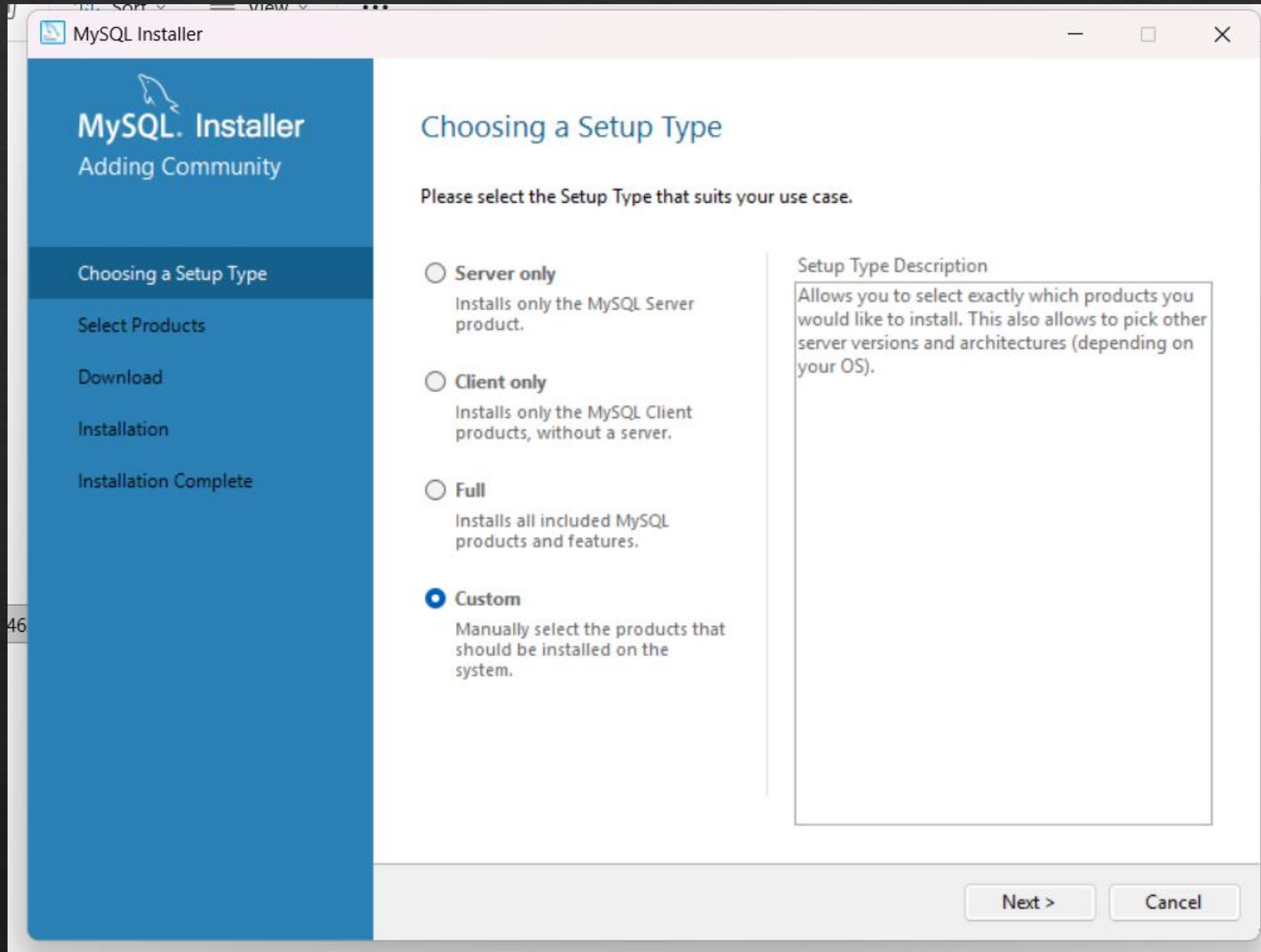
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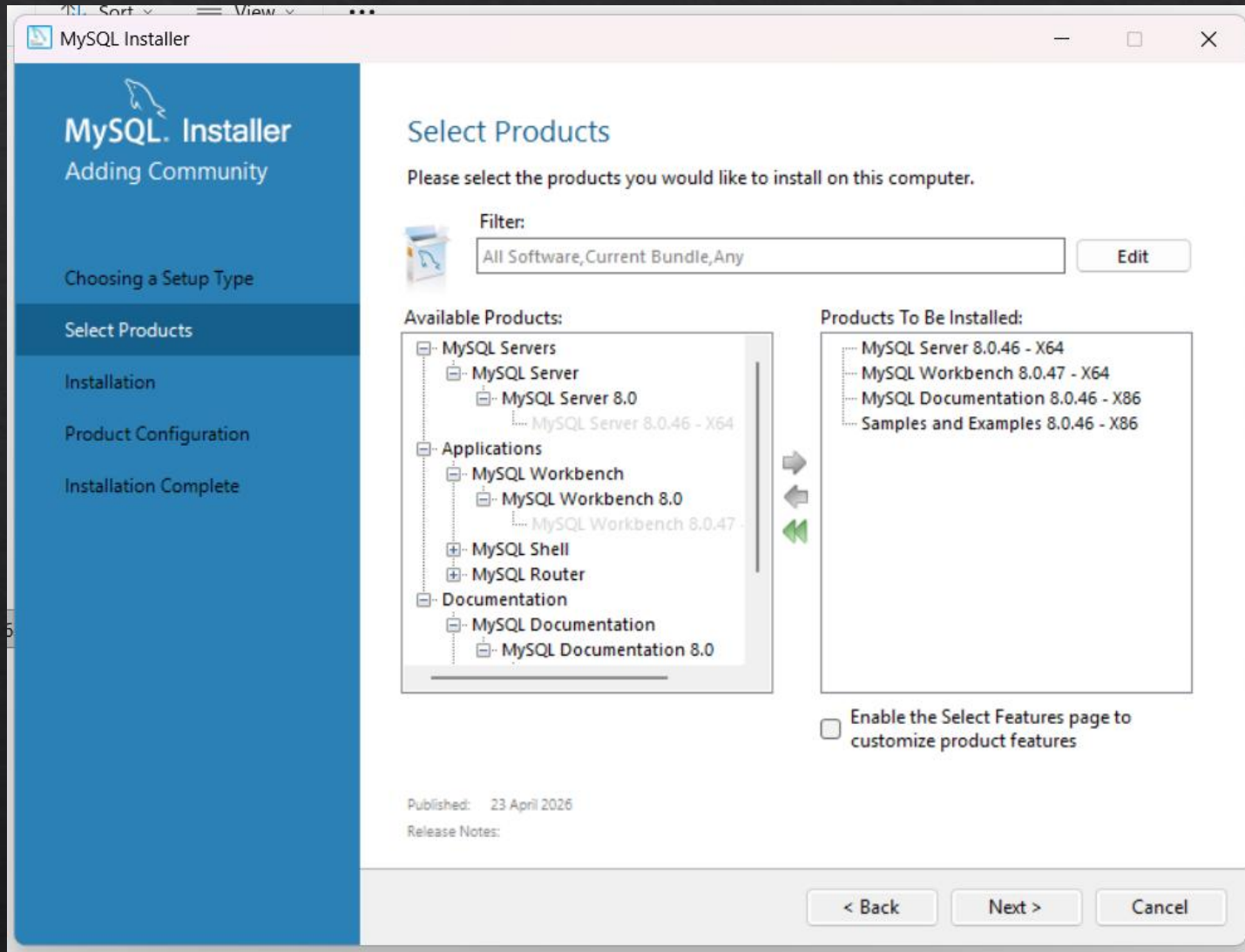


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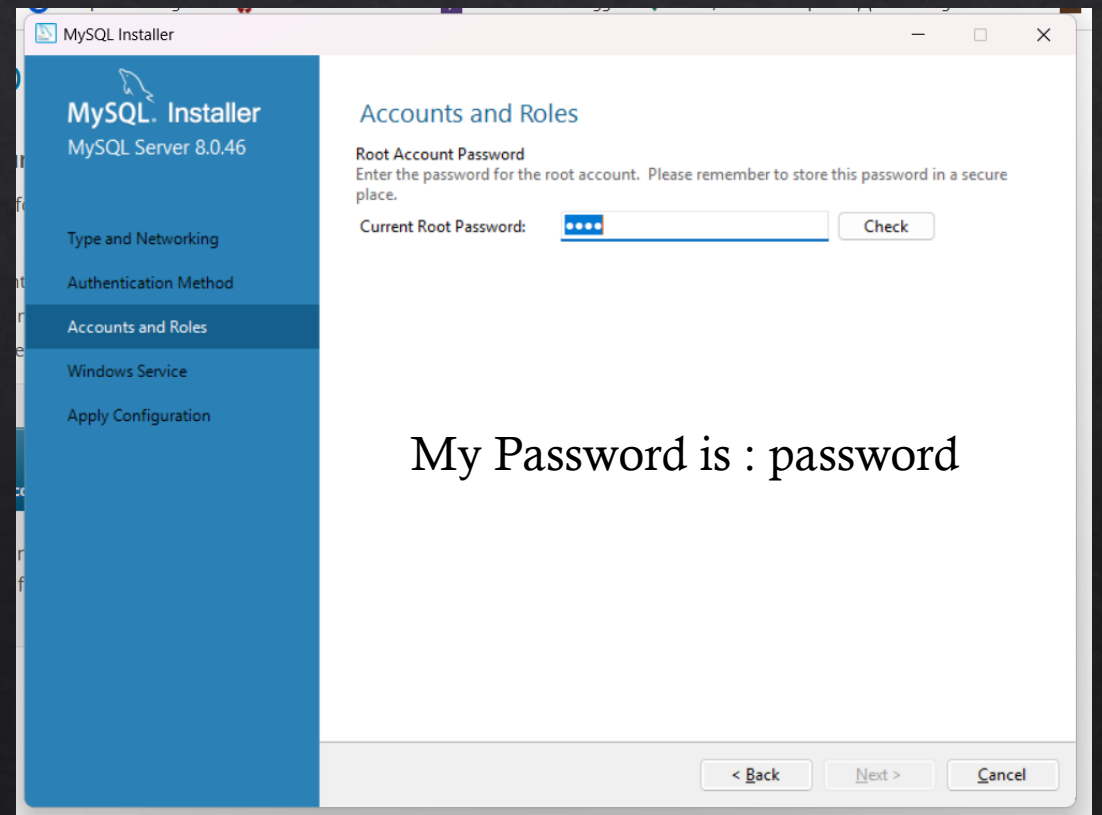
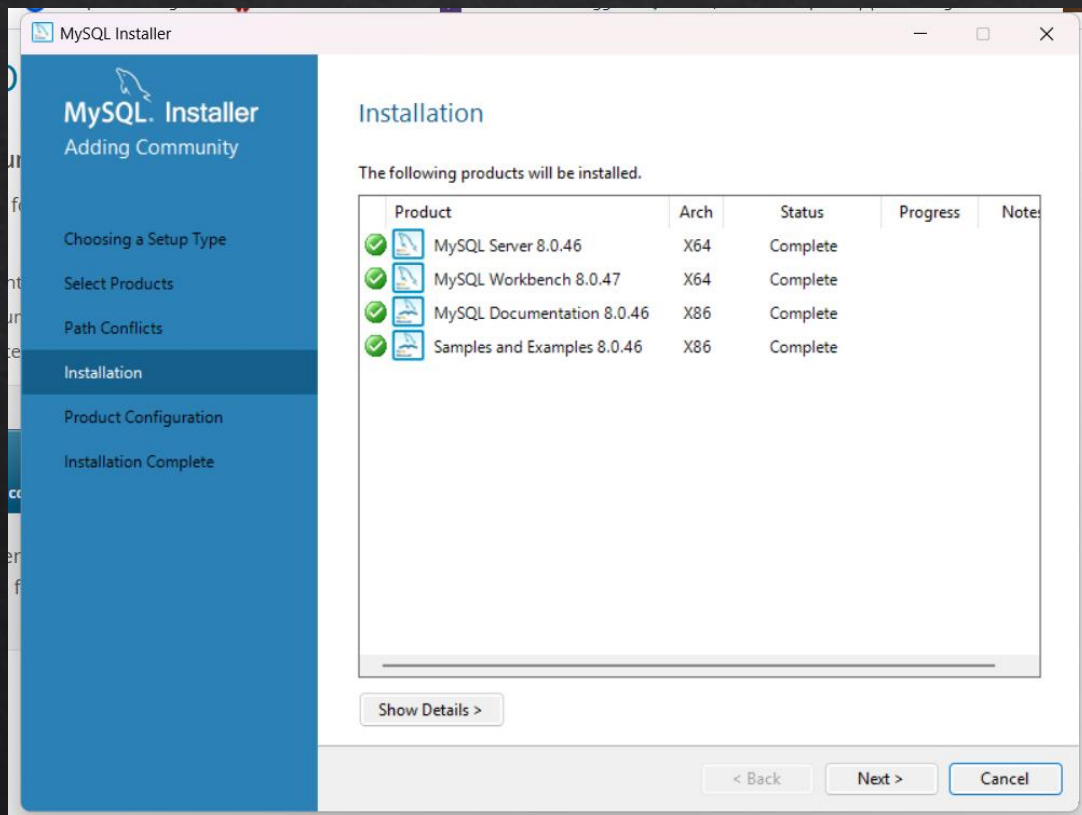
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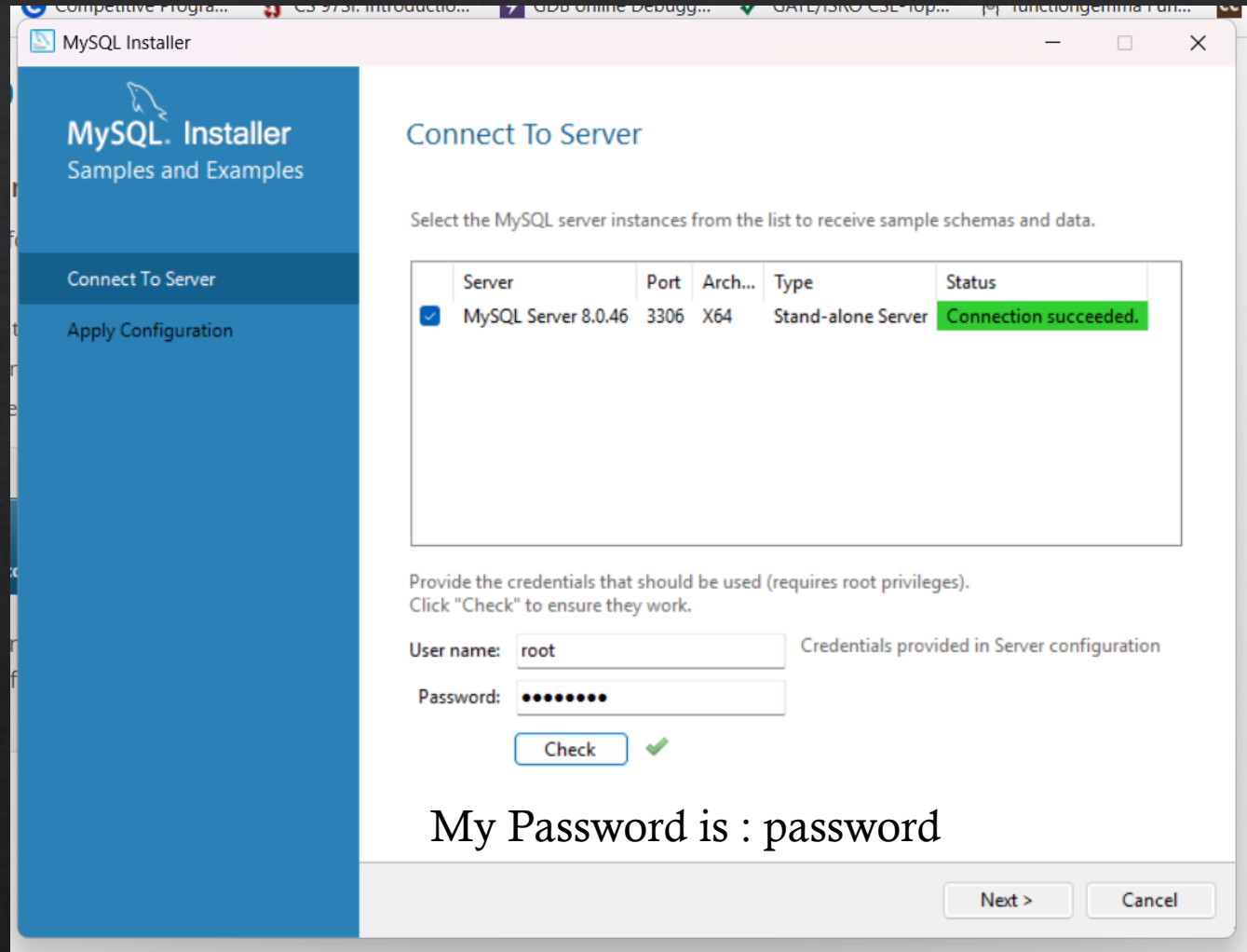
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Installing selected packages + setting root password



Connect with MySQL server



Introduction to SQL

- **SQL = Structured Query Language.**
- SQL is a language used to: Store data, Retrieve data, Update data, Delete data, Manage databases.
- SQL is mainly used with RDBMS like:MySQL, Oracle Database, PostgreSQL, Microsoft SQL Server.
- SQL keywords are not case sensitive.

SQL commands

- SQL commands are divided into different categories based on their purpose.

Category	Full Form	Purpose	Commands
DDL	Data Definition Language	Define structure	CREATE, ALTER, DROP
DML	Data Manipulation Language	Modify data	INSERT, UPDATE, DELETE
DCL	Data Control Language	Control permissions	GRANT, REVOKE
TCL	Transaction Control Language	Manage transactions	COMMIT, ROLLBACK
DQL	Data Query Language	Retrieve data	SELECT

Data Types in MySQL

- A data type defines:
 - ✓ The type of value a column can store,
 - ✓ The size of the data,
 - ✓ The valid range of values.

Category	Examples
Numeric	INT, BIGINT, FLOAT
String	CHAR, VARCHAR, TEXT
Date & Time	DATE, TIME, TIMESTAMP
Other Types	BOOLEAN, JSON

Operators in MySQL

- Operators are symbols or keywords used to:
 - ✓ perform calculations,
 - ✓ compare values,
 - ✓ filter records,
 - ✓ combine conditions.

Operator	Purpose
+ - * / %	Arithmetic
= > < >= <=	Comparison
AND OR NOT	Logical
BETWEEN	Range
IN	Multiple values
LIKE	Pattern matching
IS NULL	NULL checking
EXISTS	Subquery existence

Data Query Language (DQL)

- DQL (Data Query Language) is used to retrieve data from a database.

Command / Clause	Purpose	Query
SELECT	Retrieve data	Show all employee records from employees table
FROM	Specifies table	Show all CustomerName and City from Customer table.
WHERE	Filter rows	Show suppliers from Germany Show all details of the customer whose CustomerID is 1. Show all customer details whose CustomerID is greater than 80.
DISTINCT	Remove duplicates	Show all unique supplier countries
ORDER BY	Sort records	Show customers sorted by name Show all customer details sorted first by Country and then by CustomerName
ASC	Ascending sorting	Show orders sorted by oldest date first
DESC	Descending sorting	Show latest orders first Show all customer details sorted by Country in ascending order and CustomerName in descending order.
LIMIT	Restrict output rows	Show only first 5 customers